

Generalization at Accentuation-Optimal Regularization: A Conditional Effective-Dimension Universality

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Scope. This note concerns pixel-space ridge and the leading deterministic-equivalent (DE) formulas. The identities below are exact identities between these DE expressions, not exact finite-sample risk identities. “Universality” means that teacher dependence can be absorbed into the calibration-selected effective dimension; it does not assert a new distributional universality theorem.

1 Setup and the calibration constraint

Let $\Sigma \succeq 0$ have eigenvalues s_j , and let β_j^* be the coordinates of a fixed teacher in its eigenbasis. Assume a finite spectrum, $S = \sum_j s_j \beta_j^{*2} > 0$, and response-noise variance $\sigma^2 > 0$. The ridge objective is $n^{-1}\|y - X\beta\|^2 + \lambda\|\beta\|^2$. Define

$$B_{p,q}(\kappa) = \sum_j \frac{\beta_j^{*2} s_j^p}{(s_j + \kappa)^q}, \quad \text{df}_{p,q}(\kappa) = \sum_j \frac{s_j^p}{(s_j + \kappa)^q}, \quad \text{df}_1 = \text{df}_{1,1}, \quad \text{df}_2 = \text{df}_{2,2}.$$

On the physical branch,

$$\lambda(\kappa) = \kappa \left(1 - \frac{\text{df}_1(\kappa)}{n} \right), \quad \kappa > \kappa_0, \quad n - \text{df}_1 > 0, \quad n - \text{df}_2 > 0.$$

Here $\kappa_0 = 0$ if $\text{rank } \Sigma \leq n$; otherwise it is the positive solution of $\text{df}_1(\kappa_0) = n$. Write $V = E_{\text{gen,DE}} + \sigma^2$. The generalization and norm/overlap DEs are

$$E_{\text{gen,DE}} = \frac{n(\kappa^2 B_{1,2} + \sigma^2)}{n - \text{df}_2} - \sigma^2, \quad \mu_N = B_{1,1}, \quad \mu_D = B_{2,2} + \frac{\text{df}_{1,2}}{n} V. \quad (1)$$

Since $B_{1,1} - B_{2,2} = \kappa B_{1,2}$, the condition $z = \mu_D/\mu_N - 1 = 0$ is equivalent to

$$V = n\kappa \frac{B_{1,2}}{\text{df}_{1,2}}, \quad \sigma^2 = n\lambda(\kappa) \frac{B_{1,2}}{\text{df}_{1,2}}. \quad (2)$$

At such a point the leading quantities satisfy $E_{\text{acc}}^{(0)}/S = (z/(1+z))^2 = 0$, $(R_{\text{acc}}^2)^{(0)} = 1 - z^2 = 1$, and $\text{slope}_{\text{acc}}^{(0)} = 1/(1+z) = 1$.

For completeness, a zero exists: the right-hand side of the second equation in (??) is continuous, tends to zero as $\kappa \downarrow \kappa_0$, and tends to infinity as $\kappa \rightarrow \infty$, because $B_{1,2}/\text{df}_{1,2} \rightarrow S/\text{tr } \Sigma > 0$ and $\lambda \sim \kappa$. The zero need not be unique. This is an existence statement for the DE expressions under consideration, not a finite-sample calibration guarantee.

2 The effective-dimension risk law

Proposition 1 (Generalization at an accentuation zero). *At any calibration point $\kappa_a > \kappa_0$ with $\lambda_a = \lambda(\kappa_a)$,*

$$E_{\text{gen,DE}}(\kappa_a) = \sigma^2 \left(\frac{\kappa_a}{\lambda_a} - 1 \right) = \sigma^2 \frac{\text{df}_1(\kappa_a)}{n - \text{df}_1(\kappa_a)}. \quad (3)$$

Proof. Divide the two equations in (??) to obtain $V/\sigma^2 = \kappa_a/\lambda_a$, and then use the physical-branch relation. $\square$

A useful off-calibration identity. For any $\lambda > 0$ on the branch, direct algebra gives

$$\frac{E_{\text{gen,DE}} + \sigma^2}{\sigma^2} - \frac{\kappa}{\lambda} = -\frac{\kappa^2 B_{1,1}}{\lambda \sigma^2} z.$$

Thus the risk law is equivalent to calibration, not a formula valid at all regularizations. In particular it need not hold at a prediction optimum.

3 Why the result is structurally special

3.1 From squared shrinkage to effective degrees of freedom

Set $q_j = s_j/(s_j + \kappa)$. Then

$$\text{df}_1 = \sum_j q_j, \quad \text{df}_2 = \sum_j q_j^2, \quad \text{df}_1 - \text{df}_2 = \sum_j q_j(1 - q_j) = \kappa \text{df}_{1,2}.$$

The general risk separates into a signal-dependent contribution and a response-noise contribution:

$$E_{\text{gen,DE}} = \frac{n\kappa^2 B_{1,2}}{n - \text{df}_2} + \frac{\sigma^2 \text{df}_2}{n - \text{df}_2}. \quad (4)$$

The first term includes the effect of random design on the signal; it should not be called solely the squared bias of the mean estimator. At calibration, the teacher-dependent shrinkage term obeys

$$\kappa_a^2 B_{1,2}(\kappa_a) = \sigma^2 \frac{\text{df}_1(\kappa_a) - \text{df}_2(\kappa_a)}{n - \text{df}_1(\kappa_a)}. \quad (5)$$

Substitution into (??) yields (??). The nonnegative signal contribution effectively promotes the complexity parameter from df_2 to df_1 . Bias and variance do not cancel in the prediction MSE. Instead, calibration fixes their relation so that their sum has a single effective-dimension expression.

The difference $\sum_j q_j(1 - q_j)$ is supported on partially retained modes; it vanishes for a hard keep/discard filter. It quantifies the gap between linear and squared shrinkage. At calibration, the teacher's shrinkage loss is tied precisely to this gap and to the noise variance.

3.2 An error self-consistency relation

The general risk equation is equivalently

$$E_{\text{gen,DE}} = \kappa^2 B_{1,2} + \frac{\text{df}_2}{n} V.$$

Calibration substitutes $\kappa^2 B_{1,2} = (\text{df}_1 - \text{df}_2)V/n$, hence

$$E_{\text{gen,DE}} = \delta_a V, \quad V = \sigma^2 + \delta_a V, \quad \delta_a = \frac{\text{df}_1(\kappa_a)}{n} \in (0, 1). \quad (6)$$

Consequently $V = \sigma^2/(1 - \delta_a)$. This is an algebraic feedback interpretation of the DE, not a literal iterative training mechanism. It explains why perfect leading accentuation calibration can coexist with large prediction error when effective complexity approaches the sample budget.

3.3 OLS-like, but not an OLS equivalence theorem

Equation (??) has the familiar leading-order OLS form $\sigma^2 p_{\text{eff}}/(n - p_{\text{eff}})$ with $p_{\text{eff}} = \text{df}_1(\kappa_a)$. This does not identify ridge with a hard-truncated OLS estimator, assert unbiasedness, or give the exact finite-sample OLS expectation (whose finite-size denominators can differ). It is a risk-level structural analogy: shrinkage and estimation effects combine into an effective-complexity law at this particular constraint.

4 Weight-space geometry: calibration is radial optimality

For a fixed nonzero estimated weight, let $D = \|\hat{\beta}\|^2$ and $N = \beta^{*\top} \hat{\beta}$. Then

$$D = N \iff \hat{\beta}^\top (\beta^* - \hat{\beta}) = 0.$$

Moreover, for $L(c) = \|c\hat{\beta} - \beta^*\|^2$,

$$L'(1) = 2(D - N), \quad \arg \min_c L(c) = N/D.$$

Thus exact calibration means that the fitted weight already has the optimal global Euclidean scale along its own direction. It does not imply optimal direction or teacher recovery. The analogous expectation-level statement is $\arg \min_c \mathbb{E}L(c) = \mathbb{E}N/\mathbb{E}D$ when the moments exist; the leading DE substitutes μ_N, μ_D for these quantities.

By contrast, prediction-optimal rescaling of a fixed weight uses $\beta^{*\top} \Sigma \hat{\beta} / (\hat{\beta}^\top \Sigma \hat{\beta})$. The distinction is Euclidean versus data-covariance geometry. Even if $\beta^* = \hat{\beta} + u$ with $u \perp \hat{\beta}$, prediction error is $u^\top \Sigma u$, which need not be small.

5 What is universal, and what remains teacher-dependent?

At fixed spectrum, noise, sample size, and a shared calibration point κ_a , all teachers satisfying (??) have the same DE generalization error. The constraint fixes the relevant quadratic teacher statistic at that point. Changing the teacher generally changes the location of the calibration point, and hence $\text{df}_1(\kappa_a)$.

Accordingly the result is a *teacher-free risk expression at a teacher-dependent operating point*, not a teacher-independent numerical risk. Calibration is an oracle condition unless its needed statistics can be estimated. Finding a zero does not by itself supply a practical CV procedure.

6 Constructive oracle bound and simultaneous performance

Corollary 1. *Let $E_{\text{gen}}^* = \inf_{\lambda \geq 0} E_{\text{gen,DE}}(\kappa(\lambda))$. Every calibration point supplies a feasible witness, so*

$$0 \leq E_{\text{gen}}^* \leq \min \left\{ S, \sigma^2 \frac{\text{df}_1(\kappa_a)}{n - \text{df}_1(\kappa_a)} \right\}.$$

The S bound follows from the zero-predictor limit $\lambda \rightarrow \infty$.

The optimization inequality itself is elementary; what is constructive is the existence of a calibration solution with a computable risk and a second, simultaneously satisfied performance objective. Writing $\rho = \sigma^2/S$, one obtains

$$\frac{E_{\text{gen}}^*}{S} \leq \min\{1, \rho\delta_a/(1 - \delta_a)\}, \quad \frac{E_{\text{gen,DE}}(\kappa_a)}{S} \leq \varepsilon \iff \delta_a \leq \frac{\varepsilon}{\rho + \varepsilon}.$$

Thus simultaneous perfect leading accentuation and small prediction error require sufficiently small effective complexity relative to the noise level.

If two distinct calibration zeros satisfy $\kappa_{a,2} > \kappa_{a,1}$, then $\text{df}'_1(\kappa) = -\text{df}_{1,2} < 0$ implies $E_{\text{gen,DE}}(\kappa_{a,2}) < E_{\text{gen,DE}}(\kappa_{a,1})$. This orders risks *among calibration zeros*, not along the whole ridge path. The bound concerns the continuous DE oracle; it is not a per-trial guarantee for empirical RidgeCV or an arbitrary finite selection grid.

7 Limits of the statement

- Finite-sample ratio fluctuations remain. With $R = N/D$, $\mathbb{E}[(1 - R)^2] = (1 - \mathbb{E}R)^2 + \text{Var}(R)$. $\mu_N = \mu_D$ does not force this risk to vanish. A distributional optimum can move away from the leading calibration point.
- The displayed ratios assume $\sigma^2 > 0$ and $\lambda_a > 0$. Zero noise requires boundary limits, especially when $\text{rank } \Sigma = n$. In overparameterized problems, $\text{df}_1(\kappa_a) \rightarrow n$ can offset a vanishing noise factor, so the formula does not imply noiseless recovery.
- No monotonicity of the oracle gap or of calibration-selected effective dimension is asserted without additional spectral/alignment assumptions.
- General feature regression uses the input-space metric $FF^\top$ in the norm term. The pixel-space cancellations here do not automatically extend to a nonorthogonal feature map.

Takeaway. At leading order, zero accentuation error imposes a calibration constraint that converts the teacher-dependent shrinkage contribution into an effective-complexity contribution. Prediction risk then follows the OLS-like law $\sigma^2 p_{\text{eff}}/(n - p_{\text{eff}})$, with teacher dependence retained through the calibration-selected effective dimension.